Job-Skill Matching with Skill Type Classification*

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Abstract

This literature review addresses the task of Job-Skill Matching with Skill Type Classification. The task requires, given a job title, retrieving a ranked list of relevant skills and classifying each as core, complementary, or transversal. We are provided a training set of at least 5,000 job title–skill pairs, a development set of 200 job titles with ESCO-normalised skill annotations carrying a core/contextual relevance label, and a test set of 500 job titles for blind evaluation. We survey multiple prior works covering skill classification and skill extraction, each contributing methods or findings we draw on in building our approach.

Keywords

job-skill matching, skill type classification, ESCO, transformer models, named entity recognition, ensemble learning, information retrieval

1. Introduction to the Task

The task of Job-Skill Matching with Skill Type Classification sits at the intersection of information retrieval and multi-class labelling. A system must take a raw job title as input, retrieve a ranked list of skills from the ESCO taxonomy, and simultaneously assign each retrieved skill one of three labels: *core*, *complementary*, or *transversal*. The challenge lies in combining semantic retrieval with skill-type classification under ranking-based evaluation metrics such as nDCG and MAP. Because TalentCLEF includes multilingual data, models must operate across languages without relying solely on translation.

2. Literature Review

The work most relevant to our task spans two directions: extracting skills from raw job descriptions and classifying skills by type using taxonomy-aligned labels. The papers reviewed below each tackle a different piece of this problem, yet together they outline the methodological foundations we build upon.

The study titled "Techniques for Transversal Skill Classification and Relevant Keyword Extraction from Job Advertisements" [1] presents a two-stage classification pipeline that identifies transversal skills in job ads according to the ESCO taxonomy. The first stage places a skill into one of six broad groups—Core, Thinking, Self-management, Social/communication, Physical, and Life skills—while the second stage refines that assignment into finer subclasses. Both stages rely on 768-dimensional sentence embeddings from a BERT-based Sentence Transformer passed into neural classifiers. A particularly encouraging finding is that a multilingual embedding model came within 0.55% accuracy of an English-only model, which suggests that a separate translation step may not be necessary when handling non-English job titles in TalentCLEF. The label hierarchy used in this work maps naturally onto the core/complementary/transversal distinction required by our task.

Building on the theme of skill extraction from unstructured text, the work titled "Job Description Parsing with Explainable Transformer Based Ensemble Models to Extract the Technical and Non-Technical Skills" [2] takes an ensemble route that combines five extractors (a BiLSTM, a CNN-enhanced BiLSTM, CRF-augmented versions of both, and a standalone CRF) all operating on BERT contextualised

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embeddings, with majority voting producing the final output. The deep components are further trained under a shared multi-task objective. The author scraped around 30,000 job advertisements covering 22 IT roles, and annotated them with an inventory of over 7,000 skill surface forms. The resulting multi-task ensemble reached F-scores of 67% for non-technical skills and 72% for technical skills. Ensemble voting offers a practical way to stabilise the ranking metrics that TalentCLEF uses for evaluation, and the consistent gap between hard-skill and soft-skill performance foreshadows the difficulty of distinguishing transversal competences from core ones, given that transversal skills tend to appear in vaguer language.

Another contribution is the work titled "SkillSpan: Hard and Soft Skill Extraction from English Job Postings" [3] which introduces *JobBERT* and *JobSpanBERT*, domain-adapted BERT variants pre-trained on 3.2 million unlabelled job-posting sentences. JobBERT achieved the highest combined span-F1 of 59.73 on the test set, with Almost Stochastic Order tests confirming its dominance over vanilla BERT. Multi-task learning—predicting skill and knowledge tags jointly—did not outperform single-task training. For our project: domain-adaptive pre-training on job-posting text delivers consistent gains, which motivates a similar strategy for TalentCLEF; and skill spans, particularly for soft skills, are substantially longer than typical named entities, so any extraction component must be evaluated on multi-token spans rather than single-token accuracy.

Moving from skill extraction in job advertisements to the candidate side of the matching problem, the work titled "A Two-Phase Framework for Automated Information Extraction from Curriculum Vitae" [4] proposes an end-to-end pipeline for pulling structured fields out of CVs. The framework operates in two stages: a layout-analysis-based block segmentation step that partitions each CV into regions such as Personal Information, Education, Work Experience, and Skills, followed by a named entity recognition model that labels tokens within each region using IOB notation. The authors compare three transformer-based models—XLM-RoBERTa, BERT Multilingual Cased, and BERT Base Uncased—on a newly annotated dataset of 914 English and 400 Vietnamese CVs covering seven entity types including skill sets. XLM-RoBERTa achieved the best overall F1 of approximately 80%, though the authors note that multi-word entities such as degree names, organisation names, and job titles remain harder to detect than shorter fields like phone numbers or emails.

Shifting the perspective from span-level extraction to document-level prediction, the study titled "Retrieving Skills from Job Descriptions: A Language Model Based Extreme Multi-label Classification Framework" [5] frames skill prediction as an extreme multi-label classification (XMLC) problem over roughly 2,500 skill labels from a Singaporean government job portal. Their BERT-XMLC model encodes a job description through $\text{BERT}_{\text{BASE}}$, compresses the representation via a bottleneck layer, and predicts each skill independently through sigmoid outputs. The authors also find that over 86% of required skills are implicit—absent from the job description text and inferable only from context. This observation is relevant to us where systems must retrieve skills from a job title alone, making the ability to infer unstated skills essential.

Focusing specifically on the transversal dimension of skills, the work titled "Hierarchical Classification of Transversal Skills in Job Advertisements Based on Sentence Embeddings" [6] develops a two-level classification pipeline aligned with the ESCO transversal-skill hierarchy. At the first level, six binary classifiers determine whether a sentence refers to one of the six ESCO transversal categories; at the second level, a multi-label classifier assigns specific sub-skills within each category. All models use 768-dimensional BERT-based sentence embeddings. The authors built a dataset of 5,208 manually labelled sentences from 219 European job ads. Level-one accuracy exceeded 94% across all categories, but level-two accuracy dropped to around 68% for some sub-skills due to limited data and ambiguous boundaries. The multilingual embedding model performed within 0.55% of its English-only counterpart, reinforcing the finding from Gavrilescu et al. [1] that translation steps may be unnecessary for multilingual European data.

Finally, the work titled "Unsupervised Skill Identification from Job Ads" [7] takes a different route by avoiding labelled training data entirely. The methodology has two stages: first, job ad sentences are embedded alongside entries from the ESCO skill base using Sent2vec, and the closest ESCO skill is retrieved for each sentence via cosine similarity; second, capitalised words not captured in the first stage are looked up against Wikipedia and DBpedia to identify programming languages, software names,

and certifications. ESCO occupation clusters are then used to filter out spurious matches based on skill co-occurrence within the same ISCO-08 group. Evaluated on 100 manually labelled job ads from diverse sectors, the Sent2vec variant achieved a precision of 83% and a recall of 78% for hard skills, improving recall by 56% over a simple exact-match baseline. Notably, SBERT embeddings performed considerably worse in this setting, suggesting that off-the-shelf sentence transformers may need domain-specific fine-tuning for skill retrieval.

Table 1
Summary of reviewed literature.

Paper Title	Task	Dataset	Model(s)	Key Result
Techniques for Transversal Skill Classification [1]	Transversal skill classification	ESCO-aligned job ads	BERT Sentence Transformer + neural classifiers	Multilingual within 0.55% of English-only accuracy
Job Description Parsing with Ensemble Models [2]	Hard & soft skill extraction (NER)	30,695 scraped IT job ads; 22 roles; 7,170+ skill forms	Ensemble of BiLSTM, CNN, CRF with BERT embeddings; LIME	F1 67% (soft), 72% (hard); significant over baselines
SkillSpan: Hard and Soft Skill Extraction [3]	Span-level skill & knowledge extraction	391 JPs, 14.5K sentences, 12.5K spans	JobBERT, JobSpanBERT (domain-adapted BERT); CRF layer	Span-F1 59.73 (combined); STL > MTL
A Two-Phase Framework for CV Information Extraction [4]	CV information extraction (NER + layout segmentation)	914 English + 400 Vietnamese CVs; 7 entity types	XLM-RoBERTa, BERT Multilingual, BERT Base; CRF layer	F1 80% (XLM-RoBERTa); segmentation 97% accuracy
Retrieving Skills from Job Descriptions: BERT-XMLC [5]	Skill prediction as extreme multi-label classification	20,298 JPs; 2,548 skill labels	BERT-XMLC with bottleneck layer; CAB bootstrapping	Recall@5 +10%; nDCG@5 +7%; 86% skills implicit
Hierarchical Classification of Transversal Skills [6]	Hierarchical transversal skill classification	219 EURES job ads; 5,208 labelled sentences	Neural classifiers on BERT sentence embeddings; two-level hierarchy	Level-1 accuracy >94%; level-2 ∼68%
Unsupervised Skill Identification from Job Ads [7]	Unsupervised skill identification	100 labelled job ads; ESCO skill base	Sent2vec + ESCO similarity retrieval; Wikipedia/DBpedia	Precision 83%, Recall 78% (hard skills)

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